Sample Midterm 1, MATH 4211/6211 Optimization

NAME:____________________________**ID#:** ______________________ **Section: 4211/6211**

Show all your work. Answers without solution steps will not receive credits.

1. (4 points) Show that $\{(\boldsymbol{x}, t) \in \mathbb{R}^{n+1} : \|\boldsymbol{x}\| \leq t, t \geq 0\}$ is a convex set in $\mathbb{R}^{n+1}$ using the definition.

2. (4 points) Suppose $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_m \in \mathbb{R}^n$ and $b_1, \ldots, b_m \in \mathbb{R}$ are given. Define $f : \mathbb{R}^n \to \mathbb{R}$ as $f(\boldsymbol{x}) = -\log(-\log(\sum_{j=1}^m e^{\boldsymbol{a}_i^T \boldsymbol{x} + b_i}))$ for $\boldsymbol{x} \in \mathbb{R}^n$. Is f a convex function, and why? You can use the fact that $\log(\sum_{i=1}^n e^{x_i})$ is convex in $\boldsymbol{x} = (x_1, \ldots, x_n)^T \in \mathbb{R}^n$.

3. (4 points) Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(\boldsymbol{x}) = \boldsymbol{x}^T \begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix} \boldsymbol{x} + \boldsymbol{x}^T \begin{bmatrix} 3 \\ 5 \end{bmatrix} + 6$$

(a) Find the gradient and the Hessian of f at $\boldsymbol{x} = [1, 1]^T$.

(b) Find the directional derivative of f at $\boldsymbol{x} = [1, 1]^T$ with respect to a unit vector in the direction of maximal rate of increase.

4. (4 points) Define $f : \mathbb{R}^2 \to \mathbb{R}$ by $f(\boldsymbol{x}) = x_1 x_2$ for $\boldsymbol{x} = [x_1, x_2]^T \in \mathbb{R}^2$. Find the gradient and the Hessian of f. Is f convex and why?

5. (4 points) Suppose that $\boldsymbol{x}^*$ is a local minimizer of f over the set Ω where $\Omega \subset \Omega' \subset \mathbb{R}^n$. Show that if $\boldsymbol{x}^*$ is an interior point of Ω then $\boldsymbol{x}^*$ is a local minimizer of f over Ω'. Show that this cannot be claimed if $\boldsymbol{x}^*$ is not an interior point of Ω.